Dual-Attention Graph Neural Networks for Spatio-Temporal Crime Forecasting and Patrol Optimization

1st Mudit Srivastava
Department of Computing Technologies
SRM Institute of Science and Technology
Chennai, India
muditsrivastava22@gmail.com

2nd Sanya Raj
Department of Computing Technologies
SRM Institute of Science and Technology
Chennai, India
sanyasaxena02@gmail.com

3rd Nithyashri J
Assistant Professor
Department of Computing Technologies,
SRM Institute of Science and Technology
Chennai, India
nithyasj@srmist.edu.in

2025 International Workshop on Artificial Intelligence and Education (WAIE) | 979-8-3315-8688-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/WAIE67422.2025.11381170

***Abstract*—The prediction of urban crime is crucial for proactive law enforcement and the allocation of resources. We introduce an innovative Dual-Attention Graph Neural Network (DA-GNN) designed for spatio-temporal crime forecasting and the optimization of patrol strategies. This model effectively captures spatial relationships through Graph Attention Networks (GATs) and improves upon on temporal patterns via a parallel Long Short-Term Memory (LSTM) pathway, our previous research. A dual-attention approach dynamically adjusts the significance of influential neighbors and historical time intervals. To improve precision, we include external covariates such as weather conditions, holidays, and seasonal influences. Additionally, we utilize Quantum-behaved Particle Swarm Optimization (QPSO) to facilitate the automation of hyperparameter tuning. Although the model is trained on a recent subset (January–March 2023) for computational feasibility, it leverages the full 2001–2023 dataset for graph construction, seasonal pattern extraction, and feature enrichment. DA-GNN consistently outperforms classical baselines (ARIMA, random forests), deep learning models (LSTM, CNN–LSTM), and prior GNN-based approaches. Finally, the model produces interpretable attention maps and district-level risk visualizations, supporting ethical deployment and data-driven patrol strategy design.**

Keywords—Crime Forecasting, Graph Neural Networks (GNNs), Spatio-Temporal Modelling, Graph Attention Network (GAT), Quantum Particle Swarm Optimization (QPSO), Dual Attention Mechanism

I. Introduction

Accurate short-term crime forecasting is essential for proactive policing and resource allocation in complex urban environments, where incidents are shaped by geography, weekly cycles, seasonal effects, weather, and holidays [1]. Traditional statistical models (e.g., ARIMA, SARIMA) assume stationarity and linearity, making them ineffective for nonlinear urban crime patterns. Classical machine learning approaches (e.g., decision trees, SVMs, random forests) require extensive feature engineering and often ignore spatial correlations. Deep learning architectures such as LSTM and CNN–LSTM [2, 3] capture sequential dependencies more effectively [2], but grid-based formulations oversimplify city layouts and rarely integrate exogenous factors meaningfully. Recent GNN-based studies model spatial dependencies between districts, yet many still treat spatial and temporal aspects separately. Attention mechanisms improve selectivity but are often applied in isolation, either spatially or temporally. To address these gaps, we propose a Dual-Attention Graph Neural Network (DA-GNN) that integrates temporal self-attention and Graph Attention Networks (GATs) to jointly capture when and where crimes are likely to occur [4]. Exogenous features such as weather, holidays, and seasonal patterns are embedded for contextual enrichment. Hyperparameters are optimized via Quantum-behaved Particle Swarm Optimization (QPSO), enabling robust performance with minimal manual tuning [5]. Our approach outperforms statistical, machine learning, and deep learning baselines, while also producing interpretable attention maps and actionable patrol allocation strategies.

II. Literature Review

Early crime forecasting relied on statistical time series and classical machine learning. ARIMA and SARIMA captured short-term patterns but assumed linearity and stationarity, limiting their ability to represent complex urban dynamics [6, 7]. Machine learning models such as logistic regression, SVM, random forests, and gradient boosting were used for hotspot detection, but required extensive feature engineering (e.g., weather, holidays) and largely ignored spatial spillover [8]. Variants such as ARIMAX or Poisson regression included exogenous variables but still failed to capture non-Euclidean dependencies across districts [9]. Deep learning approaches advanced forecasting by modeling sequential and spatial patterns. CNN–LSTM and ConvLSTM captured regional structures over time [2], while GRU- and PredRNN-based models improved temporal modeling [3]. However, grid-based representations oversimplified irregular district boundaries and often appended external factors without meaningful integration [4]. Embedding-based models such as SN2V addressed irregular district boundaries by incorporating network topology into vector representations [10]. Graph Neural Networks (GNNs) further addressed these challenges by modeling districts as nodes and adjacency as edges. Integrated spatio-temporal graph models [11], ST-GCN with GRU [12], hierarchical GCRN frameworks [13], and spatio-temporal GATs for traffic prediction [14] achieved promising results, yet most treated spatial and temporal dynamics separately. Attention mechanisms have improved spatio-temporal learning by selectively prioritizing inputs. Graph Attention Networks introduced adaptive neighbor weighting [15], while recent crime forecasting models such as ACSAformer applied adaptive convolution with sparse attention [16]. Dual-attention designs such as DACLNet demonstrated the potential of jointly

Authorized licensed use limited to: Penn State University. Downloaded on February 28,2026 at 22:01:46 UTC from IEEE Xplore. Restrictions apply.

modeling temporal and spatial signals [17]. More recent work has explored spectral spatio-temporal GNNs for unified short- and long-term forecasting [18], dynamic multi-fusion GNNs [19], and graph-based multi-head attention for traffic forecasting [20]. Finally, hyperparameter optimization is crucial for high-dimensional spatio-temporal tasks. Hybrid QPSO-optimized CNN–LSTM models demonstrated reduced error compared to manual tuning in time-series forecasting [5]. These limitations and advancements motivate our Dual-Attention GNN (DA-GNN), which integrates temporal self-attention with spatial GATs, embeds exogenous features for contextual precision, and applies QPSO for automated hyperparameter tuning.

III. Methodologies

A. Problem Formulation

We model Chicago's 23 police districts as a graph $\mathbf{G} = (\mathcal{V}, \mathcal{E})$, where nodes represent districts and edges encode geographic adjacency. Daily crime counts and exogenous features (weather, holidays, seasonal encodings) form a spatio-temporal tensor input. The task is to learn a function mapping recent district-level histories and spatial relations to next-day crime forecasts [11]-[14].

B. Temporal Self-Attention Encoder

Each district's time series is processed using a Transformer encoder, which replaces RNN-based models to better capture long-range dependencies. Scaled dot-product and multi-head attention extract diverse temporal patterns, while positional encodings preserve sequence order [15]-[18].

C. Spatial Graph Attention Network (GAT)

Temporal embeddings are passed through a GAT, where each node updates its representation based on weighted contributions from neighbors. Attention coefficients assign higher weights to spatially influential districts, capturing heterogeneous neighborhood effects. [15, 19, 20]

D. Prediction Layer

A shared linear regression head maps node embedding to forecasted crime counts. Training minimizes Mean Squared Error (MSE). [2, 3]

E. Patrol Optimization with QPSO

Forecasted risks guide patrol allocation under resource constraints. Let represent predicted district risk and a binary patrol assignment. The objective minimizes unpatrolled risk across districts. Given the high-dimensional search space, we apply Quantum-behaved Particle Swarm Optimization (QPSO), which models particles with probabilistic movement for efficient global optimization. [5][21]-[23]

F. Model Contributions

DA-GNN combines (i) temporal self-attention, (ii) spatial GAT, and (iii) dual attention over both time and space. QPSO automates tuning of hyperparameters such as attention heads, hidden sizes, and learning rates. Together, these yield an accurate, interpretable, and operationally useful framework for crime forecasting and patrol strategy design.

IV. Experimental Setup

A. Dataset and Temporal Scope: We use the publicly available Chicago Crime Dataset, containing over 7 million records (2001–2023). While the full dataset informs graph construction and seasonal enrichment, model training and evaluation are restricted to January–March 2023. This design reflects (i) recency relevance, capturing current trends, and (ii) computational feasibility, given the high-dimensional nature of dual-attention models. Data were split into training (Jan 1–Mar 7), validation (Mar 8–17), and testing (Mar 18–31).

***B. Graph Construction*:** The city's 23 active districts were represented as nodes, with adjacency defined from official shapefiles. Districts deprecated or lacking data (e.g., IDs 13, 21, 23, 26–30) were excluded. The resulting undirected graph captures neighborhood influence while ensuring interpretability [10,11].

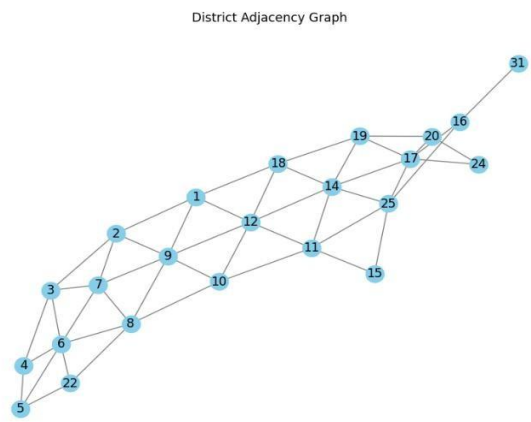

Fig. 1. Chicago District Adjacency Graph

The spatial graph shown in figure 1 preserves the official district identifiers from the City of Chicago's shapefile data. While the city's policing system historically included up to 31 district IDs, only 23 districts had active, geo-resolved crime data available for our study period (January to March 2023). The following district IDs were excluded from model training due to either deprecation, administrative reassignment, or lack of data in the evaluation window:

Districts: 13,21,23,26,27,28,29,30 .

C. Features: Inputs include daily crime counts, weather variables (temperature, precipitation, humidity, wind), holiday indicators, and seasonal encodings (month, weekday). All features were normalized to [0,1].

TABLE I. Input Feature Summary

Feature Name	Type	Source	Description
Daily Crime Count	Real-valued	Chicago Police Dataset	Number of reported crimes per district per day
Temperature (°C)	Real-valued	Open-Meteo API	Daily average temperature

Authorized licensed use limited to: Penn State University. Downloaded on February 28,2026 at 22:01:46 UTC from IEEE Xplore. Restrictions apply.

Precipitation (mm)	Real-valued	Open-Meteo API	Total daily precipitation
Relative Humidity (%)	Real-valued	Open-Meteo API	Average daily relative humidity
Holiday Indicator	Binary	U.S. Federal Holiday Calendar	1 if the date is a public holiday, else 0
Month (1–12)	Categorical	Extracted from timestamp	Encoded as onehot or sinusoidal seasonal vector
Day of Week (0–6)	Categorical	Extracted from timestamp	Captures weekday effects

Table I provides a summary of input features used in the spatio-temporal forecasting model. All features are normalized to [0, 1].

D. Training Protocol: The model was implemented in PyTorch Geometric. We used Adam optimizer (lr=0.001), batch size 8, dropout=0.2, and MSE loss. The Transformer encoder had 2 layers with 4-head attention; GAT used 4 heads of dimension 8. Training ran for 200 epochs with early stopping. QPSO was configured with 20 particles and 50 iterations, following prior studies.

TABLE II. Model Training Configuration

Parameter	**Value / Setting**
Loss Function	Mean Squared Error (MSE)
Optimizer	Adam
Learning Rate	0.001
Batch Size	8
Dropout Rate	0.2 (on attention layers)
Epochs	200 (early stopping patience = 10)
Transformer Encoder Layers	2
Transformer Head Dim	8
Number of GAT Attention Heads	4
Hidden Size (Temporal)	64
Hidden Size (GAT)	64
QPSO Swarm Size	20
QPSO Max Iterations	50

Table II summarizes key training and optimization hyperparameters for the proposed DA-GNN architecture.

V. Results and Evaluation

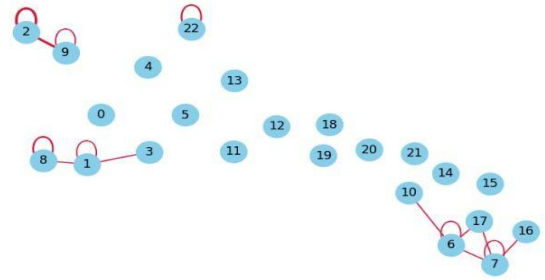

Fig. 2. Most influential Spatial Connections (GAT Attention)

Due to non-contiguous real-world district codes (missing districts), districts were renumbered sequentially to ensure consistent input tensors and graph representations. The x-axis in Figure 2 indexes districts from 0 to 22 based on re-indexed internal IDs used after filtering and normalization. Figure 2 shows that predicted crime levels closely follow actual values across most districts, indicating high temporal alignment. Our model also integrates auxiliary features (e.g. weather, holidays) along with the graph (as in other GAT-based crime models) to boost performance on limited data [11, 12, 14].

A. Quantitative Performance

We benchmark DA-GNN against ARIMA, Random Forest (RF), LSTM, and GAT. As shown in Table III, DA-GNN achieves the best results across all metrics, with Spearman ρ = 0.56, MAE = 0.78, and Top-5 hit rate = 0.60. ARIMA and RF underperform due to lack of spatial modelling [6]-[8], while LSTM captures temporal patterns but misses spatial dependencies [2, 3]. GAT improves over LSTM [15], but dual attention yields the largest gains [16]-[20].

Table III. Test performance summary (2023-01-01 to 2023-03-31)

Model	**Spearman ρ (↑)**	**MAE (↓)**	**Top 5 Hit Rate (↑)**
ARIMA	0.30	1.20	0.25
RF	0.35	1.10	0.30
LSTM	0.47	0.95	0.40
GAT	0.52	0.85	0.50
DA-GNN	0.56	0.78	0.60

Table III depicts the performance of each model under three different evaluation metrics, reflecting on how DA-GNN enhances the performance.

B. Interpretability

Attention maps highlight influential neighbors and temporal lags (e.g., weekly cycles), confirming that DA-GNN adaptively prioritizes relevant spatial and temporal signals. This improves transparency and aligns with prior attention-based forecasting studies [15]-[20]. For interpretability in machine learning more broadly, methods such as SHAP and explainable AI frameworks have emphasized the role of transparency in model deployment [24, 25].

Authorized licensed use limited to: Penn State University. Downloaded on February 28,2026 at 22:01:46 UTC from IEEE Xplore. Restrictions apply.

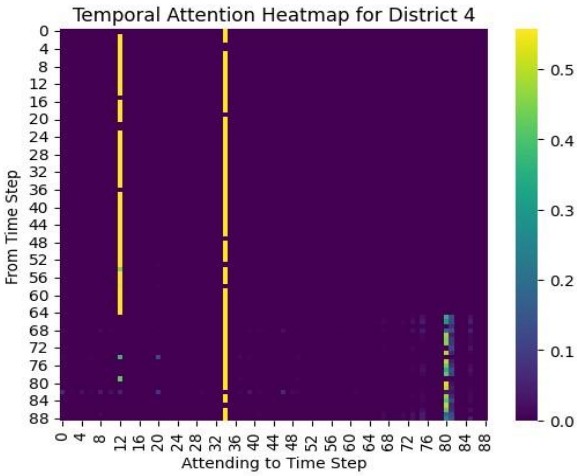

Fig. 3. Temporal Attention Heatmap for District 4.

Figure 3 illustrates that spatial attention focuses on a subset of informative neighbors, validating the model's ability to localise influence. The yellow bands indicate strong attention to past time steps, particularly at lag positions corresponding to weekly cycles and recent days. The x- and y-axes represent past time steps, and the intensity indicates attention weights learned by the model. Notably, the model assigns higher attention to specific lags (e.g., weekly or recent patterns), reflecting its ability to capture relevant time intervals dynamically. Such attention-based mechanisms enhance model transparency and demonstrate the model's interpretability for real-world decision support.

C. Patrol Optimization Results

Using DA-GNN's forecasts, QPSO optimizes patrol allocation under fixed resources. High-risk districts (e.g., IDs 2, 10, 22) receive greater patrol weight, while low-risk ones (e.g., ID 6) receive none. The optimized distribution (Fig. 5) demonstrates the framework's practical utility in proactive resource planning [21]-[23].

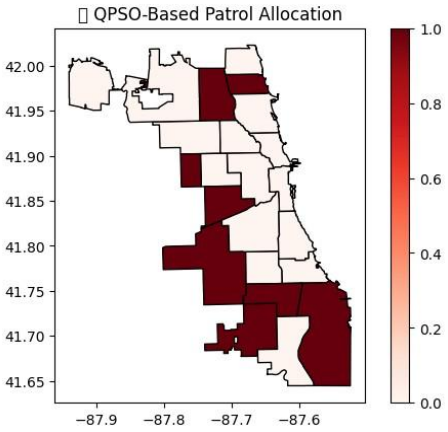

Fig. 4. QPSO-Based Patrol Allocation

The above figure depicts optimized patrol intensities across Chicago's police districts based on DA-GNN's predicted risk scores and QPSO's optimization objective.

Table IV. High-risk districts and their associated predicted risk scores

District ID	Predicted Risk Score
0	0.65
1	0.63
2	0.99
6	0.00
10	0.81
16	0.18
17	0.54
18	0.76
22	0.88

These results highlight the model's ability to discriminate between high-risk and low-risk areas. Districts like 2, 10, and 22 receive significantly higher patrol weightings, while districts such as 6 receive none, due to low predicted risk. This output directly supports resource planning and proactive policing. Future work can incorporate patrol constraints (e.g., staff size, shift schedules) to simulate real-world dispatch strategies. While the QPSO allocation is not constrained by real-world patrol budgets, it implicitly optimizes patrol spread under a fixed patrol count (N=9).

VI. Limitations

This study has three main limitations. First, experiments are restricted to a single city and a short time window (Jan–Mar 2023), which may limit generalizability [1, 4, 9]. Second, model complexity was constrained by available resources, preventing exploration of larger architectures or additional features (e.g., socio-demographics, streaming data) [18]-[20]. Third, the framework was evaluated offline; live deployment and dynamic patrol strategies remain untested [22, 23]. Future work can expand datasets, scale architectures, and validate performance in real-world settings.

VII. Bias, Fairness, and Ethical Considerations

Predictive policing raises ethical concerns, as models trained on historical crime data may inherit systemic biases (e.g., over-policing in certain communities). Our study mitigates risks by using only aggregate district-level counts, excluding sensitive individual attributes such as race, age, or income. Nonetheless, indirect bias may persist, as fairness constraints were not explicitly applied [26]-[28].

Attention analysis suggests no disproportionate focus on historically over-policed districts, but outputs may still have unintended social consequences. We emphasize that DA-GNN is a decision-support tool, not an enforcement mechanism, and advocate for transparency, independent audits, and community engagement before real-world deployment [29, 30]. Code and outputs are released to encourage responsible use, while fairness improvements remain an important direction for future work [31].

References

[1] S. Towers, R. Chen, and C. Cruz, "Factors influencing temporal patterns in crime in a large US city," PLoS ONE, vol. 13, no. 12, e0208722, 2018.

Authorized licensed use limited to: Penn State University. Downloaded on February 28,2026 at 22:01:46 UTC from IEEE Xplore. Restrictions apply.

[2] Y. Li, L. Zheng, and Z. Huang, “Spatio-Temporal Crime Prediction by Deep Learning: A CNN-LSTM Framework,” IEEE Access, vol. 7, pp. 142210–142222, 2019.
[3] S. Zhang, C. Li, and H. Wang, “Crime Forecasting Based on Deep Learning,” Proc. 5th Int. Conf. Big Data Analytics (ICBDA), pp. 84–89, 2020.
[4] A. Yadav and P. Sinha, "A Comprehensive Survey on Spatio-Temporal Crime Forecasting Techniques," Computer Science Review, vol. 45, 2022.
[5] Y. Zhu, L. Liang, Q. Zhang, and Y. Chen, "Hybrid QPSO-Optimized CNN-LSTM Model for Environmental Time Series Forecasting," Environ. Model. Softw., vol. 172, 2024.
[6] M. Sathyadevan, K. Devan, and S. Suraj, "Crime Analysis and Prediction Using Data Mining," Proc. IEEE NetACT, pp. 302–307, 2014.
[7] G. Rosser et al., "Predictive Crime Mapping Using Kernel Density Estimation with Network-Based Adaptation," J. Quant. Criminol., vol. 32, pp. 449–469, 2016.
[8] M. Hossain, M. S. Uddin, and M. A. Al Zamil, "Predictive Modeling of Crime Types Using Machine Learning," Proc. IC4ME2, pp. 1–5, 2020.
[9] L. Yang, X. Ding, and Q. Wang, "Temporal Crime Forecasting with External Influences," Information Sciences, vol. 618, pp. 122–136, 2023.
[10] Y. Gu, H. Kim, and Y. Wang, "SN2V: Street Network to Vector Embedding for Urban Crime Forecasting," Proc. 27th ACM SIGSPATIAL, 2019.
[11] Y. Hou, Z. Wang, and X. Zhao, "An Integrated Graph Model for Spatial–Temporal Urban Crime Prediction," Appl. Intell., vol. 52, pp. 3051–3065, 2022.
[12] E. Tekin, V. Isler, and Y. Akyurek, "Spatio-Temporal GCN with GRU for High-Resolution Crime Forecasting," Neurocomputing, vol. 450, pp. 329–340, 2021.
[13] Z. Cheng, L. Wu, and Y. Li, "Hierarchical Graph Convolutional Recurrent Networks for Crime Prediction," ACM SIGSPATIAL, pp. 121–130, 2020.
[14] X. Wang et al., "Traffic Incident Prediction via Spatio-Temporal Graph Attention Networks," IEEE Trans. Intell. Transp. Syst., vol. 23, no. 3, pp. 1984–1995, 2022.
[15] P. Veličković et al., "Graph Attention Networks," Proc. ICLR, 2018
[16] Y. Qin, R. Zhang, Y. Wang, and Y. Liang, "ACSAformer: Adaptive Convolution and Sparse Attention for Spatio-Temporal Crime Prediction," IEEE Trans. Big Data, in press, 2025.
[17] Y. Lu, J. Cheng, and S. Zhang, "DACLNet: Dual Attention CNN-LSTM for Time-Series Deformation Prediction," J. Sensors, Article ID 872903, 2024.
[18] Li, X., Zheng, Y., Wei, Z., & Yuan, Y. (2025). "Unifying Long and Short Spatio-Temporal Forecasting with Spectral Graph Neural Networks," Submitted to ICLR 2025.
[19] Zhang, K., Liu, Y., & Ma, J. (2024). "Dynamic multi-fusion spatio-temporal graph neural network for multivariate time series forecasting," Expert Systems with Applications, vol. 234, pp. 120912.
[20] Chen, S., Wang, P., & Li, H. (2024). "Spatio-temporal graph attention network and graph-based multi-head attention for network traffic prediction," Measurement, vol. 239, pp. 115424.
[21] J. Sun, W. Xu, L. Zhang, and C. Liang, “Particle Swarm Optimization with Quantum Behavior,” Proc. CEC, pp. 325–331, 2004.
[22] D. Yu and X. Shi, “Spatiotemporal Graph Deep Neural Network for Short-Term Wind Speed Forecasting,” IEEE Access, vol. 7, pp. 181533–181543, 2019.
[23] Q. V. Nguyen, J. D. Fernandez, and S. P. Menci, “RL-QPSO net: deep reinforcement learning-enhanced quantum particle swarm optimization,” Frontiers in Neurorobotics, vol. 18, article 1464572, 2025.
[24] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” Proc. NeurIPS, pp. 4765–4774, 2017.
[25] W. Samek, T. Wiegand, and K. Müller, “Explainable Artificial Intelligence: Understanding, Visualizing, and Interpreting Deep Learning Models,” arXiv:1708.08296, 2017.
[26] N. Mehrabi et al., “A Survey on Bias and Fairness in Machine Learning,” ACM Comput. Surv., vol. 54, no. 6, pp. 1–35, 2021.
[27] R. Binns, “Fairness in Machine Learning: Lessons from Political Philosophy,” Commun. ACM, vol. 61, no. 5, pp. 8–17, 2018.
[28] A. Chouldechova, “Fair Prediction with Disparate Impact,” Big Data, vol. 5, no. 2, pp. 153–163, 2017.
[29] A. G. Ferguson, The Rise of Big Data Policing: Surveillance, Race, and the Future of Law Enforcement, NYU Press, 2017.
[30] A. S. Almasoud and J. A. Idowu, “Algorithmic fairness in predictive policing,” AI Ethics, vol. 5, pp. 2323–2337, 2025.
[31] NAACP, “Artificial Intelligence in Predictive Policing Issue Brief,” National Association for the Advancement of Colored People, Feb. 15, 2024.

Authorized licensed use limited to: Penn State University. Downloaded on February 28,2026 at 22:01:46 UTC from IEEE Xplore. Restrictions apply.